DSA0605 – DATA HANDLING AND VISUALIZATION

CO1 – SCENARIO-BASED ASSIGNMENT

Monsoon Visual Intelligence: Mapping Rainfall Data onto Aesthetics, Axes and Color for Disaster Communication

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Course: DSA06 – Data Handling and Visualization | Tool: R with ggplot2

Dataset note: Synthetic demonstration data are used because no official faculty/IMD CSV was attached. Do not use these values for real disaster decisions.

Introduction and Dataset

The State Disaster Management Authority (SDMA) monitors rainfall, departure from normal, IMD category, reservoir storage and inflow, wind, evacuation, district, region, coastal status and date across 38 Tamil Nadu districts and five Northeast Monsoon seasons. This report follows the six stages and deliverables in the uploaded assignment.

The accompanying synthetic rainfall dataset contains 17,480 district-day records (38 districts × 92 days × 5 seasons). Separate illustrative files contain five-reservoir inflow/storage, cyclone wind direction/speed and evacuation counts. Replace them with official data before operational use.

Dataset	Records	Fields
Rainfall	17,480	Season, date, district, region, coastal, actual/normal mm, departure %, IMD category
Reservoir	2,300	Season, date, five reservoirs, inflow, storage
Wind	240	Direction degrees, speed km/h
Evacuation	38	District, people evacuated

Stage 1 – Interpret Data and Visualization Requirements (10 marks)

1(a) Visual questions and audiences

Question	Recommended graphic	Audience / purpose
Which districts have excess/deficient rainfall?	Sorted departure chart and district summary	Collectors: identify districts above/below normal.
How does reservoir inflow change and when are peaks?	Reservoir time-series line chart	Control room: monitor flood risk and rapid inflow.
From which direction do strongest winds arrive?	Polar wind rose with speed bands	Control room: cyclone direction/intensity.
Which coastal districts need immediate attention?	Red-alert highlight chart and table	Control room and Collectors: prioritize response.
How do five seasons compare?	Small multiples with common scales	Planners/public: compare seasonal patterns.

The control room needs detailed and timely information. District Collectors need clear district-level summaries. The public needs simple labels, a short legend, mobile-friendly graphics and plain-language explanations.

1(b) Variable classification and aesthetics

Variable	Type	Suitable aesthetic	Unsuitable / reason
Actual rainfall (mm)	Continuous	x/y position; sequential color	Shape is poor for quantitative magnitude.
Normal rainfall (mm)	Continuous	x/y position	Unordered hue does not express magnitude.

Departure (%)	Continuous, signed	Position; diverging color centred at 0	Sequential scale hides positive/negative meaning.
IMD category	Ordinal	Ordered color/fill	Random hues ignore natural order.
Storage (%)	Continuous, bounded	Position or sequential color bar	Categorical colors imply groups, not magnitude.
Inflow (cusecs)	Continuous, skewed	y position with log/sqrt transform	Linear axis compresses low values.
Wind speed	Continuous	Length/radial distance or intensity	Shape is not precise for numeric values.
Wind direction	Circular	Angle in polar coordinates	Linear axis breaks 0°/360° continuity.
Evacuated people	Discrete count	Area-scaled point size	Radius-proportional points exaggerate.
Relief camps	Discrete count	Position or labels	Continuous gradient may imply fractional counts.
District	Nominal, 38 levels	Labels, sorted axis, facets	38 colors/shapes are not distinguishable.
Region	Nominal, 4 levels	Qualitative color hue	Sequential scale implies order.
Coastal/inland	Binary nominal	Highlight color or shape	Continuous gradient unnecessary.
Date	Date-time	x-axis	Shape/fill does not preserve time order.

Perceptual accuracy generally follows position > length > area/angle > color intensity. Use position for quantitative comparison, hue for a small number of categories, and labels/facets for many districts.

1(c) Range, skewness and display constraints

- Reservoir inflow ranges from below 10 to above 100,000 cusecs; log scaling reveals several orders of magnitude. Since log(0) is undefined, use log1p(inflow) or a stated pseudo-log transformation.
- Departure ranges from −100% to above +300%; use a diverging scale centred exactly at zero.
- Evacuated population is likely right-skewed; use area-scaled points, not radius-scaled points.
- For color-vision deficiency use viridis/ColorBrewer; for grayscale add shape, labels and line types; for mobile reduce clutter and enlarge text.

Visualization Requirement Statement: Build a consistent and accessible R/ggplot2 dashboard for 38 districts and five seasons. Use position for quantitative comparisons, qualitative hue for four regions, ordered colors for IMD category, a zero-centred diverging scale for departure, sequential viridis for storage, log1p for skewed inflow, polar coordinates for wind direction and area-scaled size for evacuation. All graphics need units, readable labels, meaningful legends and source captions.

Stage 2 – Evaluate Aesthetic Mappings and Tool (15 marks)

2(a) R ggplot2 compared with Python

Criterion	R / ggplot2	Python / Matplotlib / Seaborn
Mapping	aes() maps x, y, color, fill, size, shape, alpha.	Matplotlib explicit arguments; Seaborn hue, size, style.
Scales/legends	Scale functions and automatic legends.	Flexible normalization, colormaps and legends; often more manual.
Layering/facets	Grammar-of-graphics layers and facet_wrap().	Subplots and Seaborn FacetGrid.
Reusability	Themes and functions support consistent dashboard.	Functions/styles support detailed customization.
Decision	Selected for concise mapping and consistent scales.	Good alternative for highly customized Python workflows.

2(b) Three scatter plot mappings

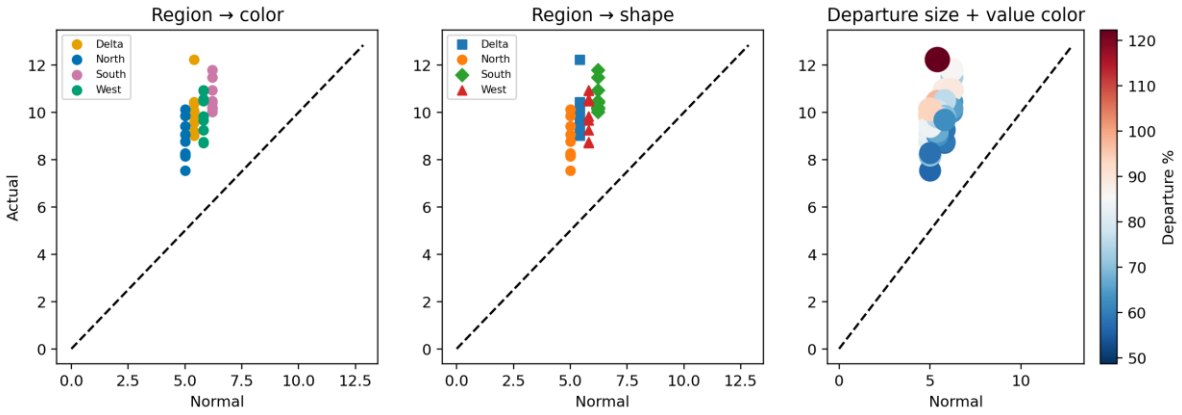


Figure 1. Three mappings: region-to-color; region-to-shape; departure-to-size with value color.

Mapping	Advantage	Limitation / evaluation
Region → color	Four regions are quickly distinguished.	Needs CVD-safe palette; best primary grouping.
Region → shape	Works in grayscale.	Dense points and small screens make shapes harder to distinguish.
Departure → size; category → color	Displays multiple variables.	Area is imprecise; signed departure cannot directly be mapped to ordinary positive size. Prefer position/diverging color for departure.

Position is more accurate than area or color intensity. Region-to-color is the clearest primary mapping for four regions; shape can be a redundant grayscale cue.

2(c) Design Decision 1

I recommend R with ggplot2, mapping normal rainfall to x, actual rainfall to y, region to color and evacuated population to point area. The 45° line shows equality; above it indicates actual rainfall above normal and below it indicates deficiency. Set2/viridis and explicit legends improve accessibility and consistency.

Stage 3 – Scales, Coordinates and Axes (15 marks)

3(a) Daily rainfall and zoom

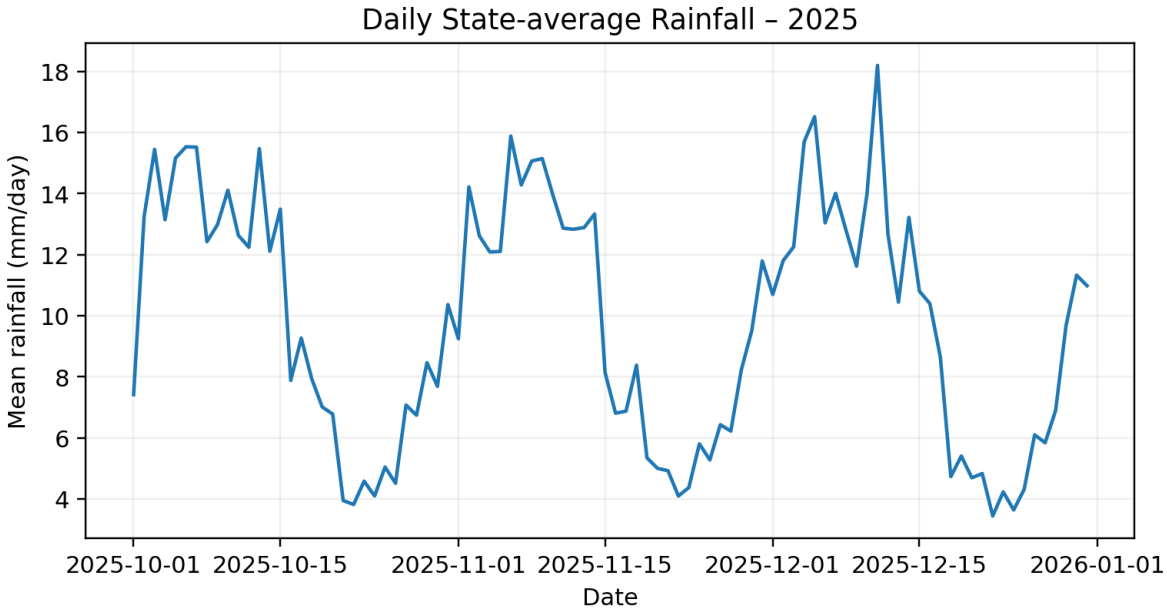


Figure 2. Daily state-average rainfall in 2025.

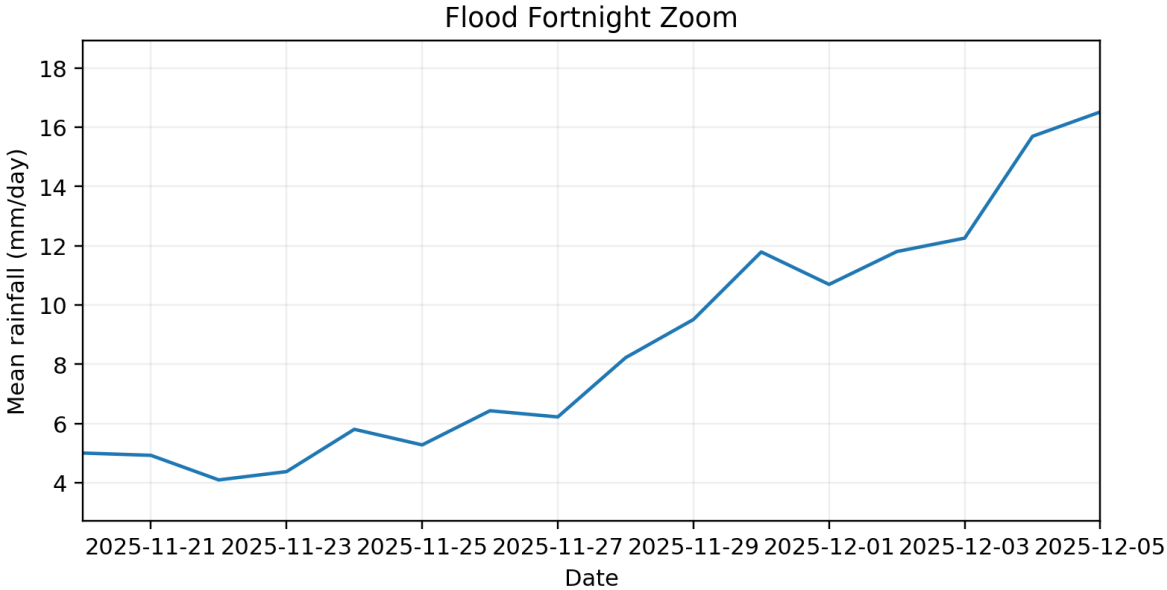


Figure 3. Zoomed flood fortnight, 20 Nov–5 Dec 2025.

Use a Cartesian date axis with custom date breaks and rainfall in mm/day. coord_cartesian() changes the visible window but retains all observations for statistical calculations. Scale limits/filtering remove data and may change summaries or smoothers. Therefore use coord_cartesian() for visual zoom.

3(b) Reservoir transformations and evacuation size

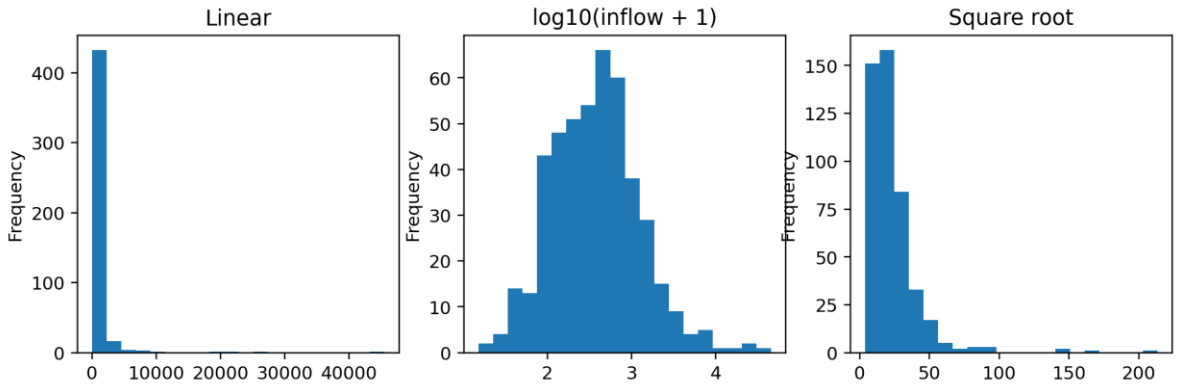


Figure 4. Linear, log10(inflow + 1) and square-root views.

A linear axis preserves additive differences but extreme values compress small inflows. A log axis shows multiplicative change; zero is undefined, so log1p is used and labelled as inflow + 1. Square root is an intermediate transformation and permits zero. Use scale_size_area() for evacuation so symbol area, not radius, is proportional to the count.

3(c) Wind rose, pie chart and equal-aspect plot

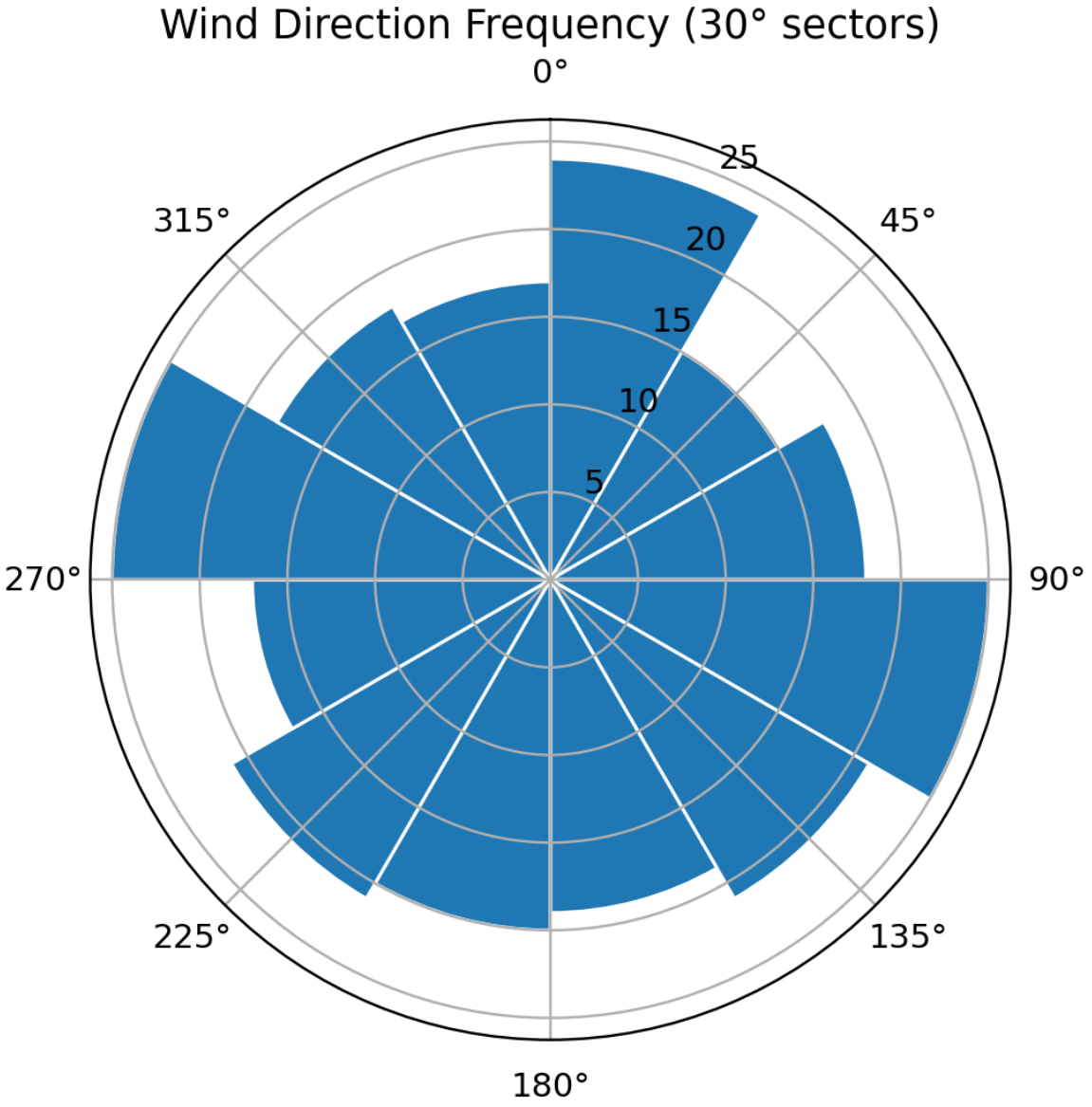


Figure 5. Wind rose with 30-degree sectors.

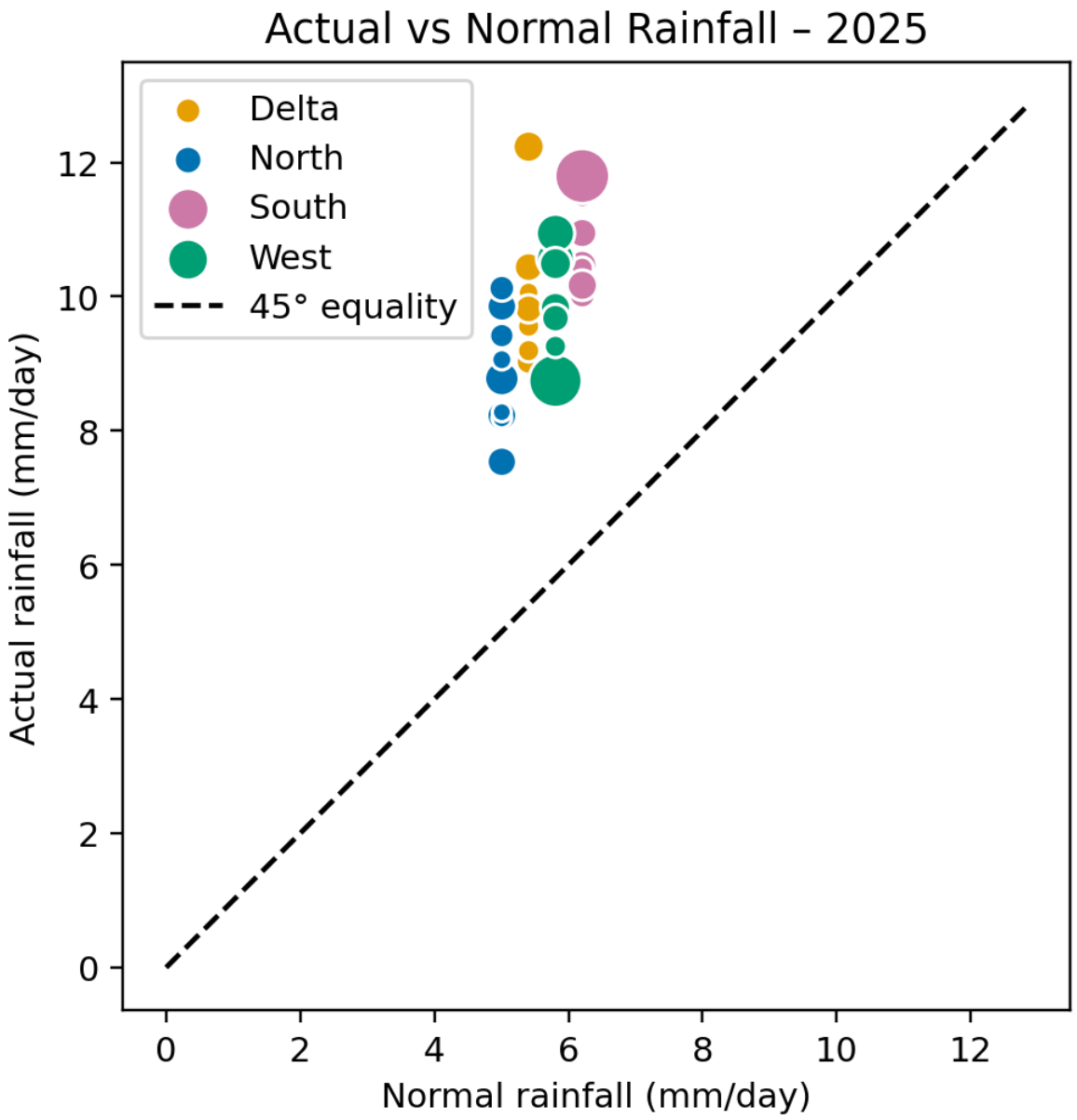


Figure 6. Actual vs normal rainfall with equal aspect and 45° line.

Wind direction is circular; polar coordinates preserve continuity between 0° and 360°. A Cartesian plot introduces an artificial boundary. A pie chart is produced by applying coord_polar(theta='y') to a stacked bar, but an ordinary bar chart compares category sizes more accurately because lengths are easier to judge than angles.

Graphic	Coordinates	Scale / transformation	Reason
Daily rainfall	Cartesian	Date; linear mm/day	Time and magnitude.
Flood zoom	Cartesian + coord_cartesian	Same scale, selected dates	Zoom without deleting observations.
Reservoir inflow	Cartesian	Linear, log1p, square root	Wide range; log1p handles zero.
Evacuated population	Cartesian	Area-scaled size	Area proportional to count.

Wind rose	Polar	Angle = direction; radius/fill = frequency/speed	Circular direction.
Category pie	Polar transformation	Proportion of total	Part-to-whole, but bar is more accurate.
Actual vs normal	Cartesian, equal aspect	Same units; 45° line	Fair equality comparison.

Design Decision 2

I recommend log1p for reservoir inflow and polar coordinates for wind direction. Log1p shows low and extreme values while allowing zero; polar coordinates correctly represent circular direction. Label the transformation and retain a linear companion for public communication.

Stage 4 – Color Scales and Palettes (20 marks)

4(a) Distinguish regions and represent storage

Use Set2/Dark2 as a qualitative palette for four regions. Thirty-eight district colors would be difficult to distinguish; use sorted labels, facets or direct annotations. Use sequential viridis for storage percentage because it represents ordered quantitative values.

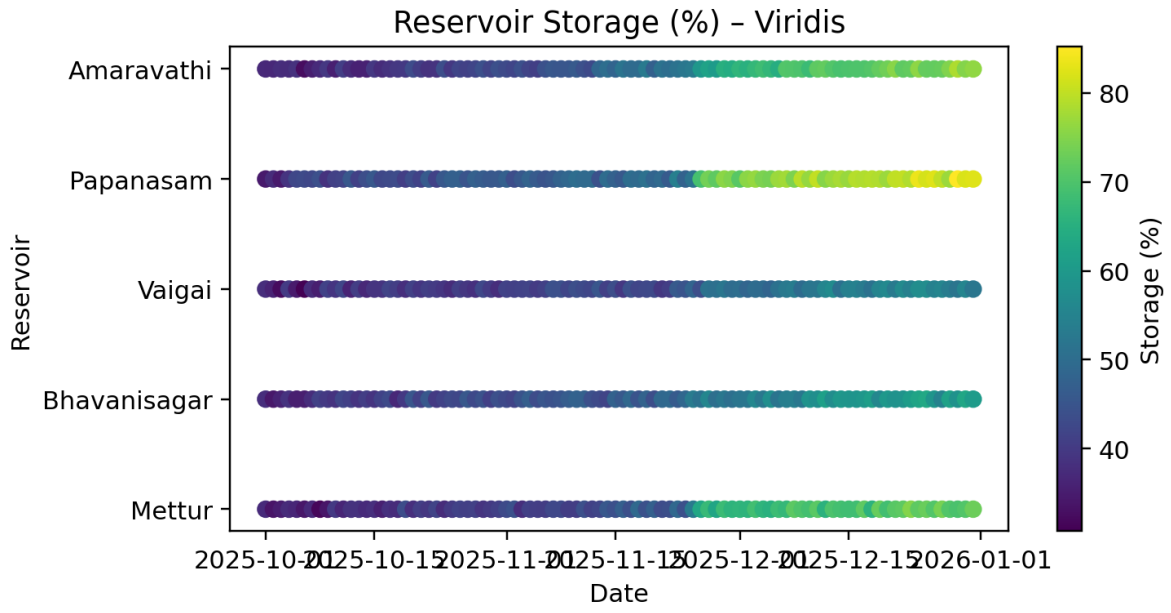


Figure 7. Reservoir storage with a sequential viridis color bar.

4(b) Diverging, ordered and highlight scales

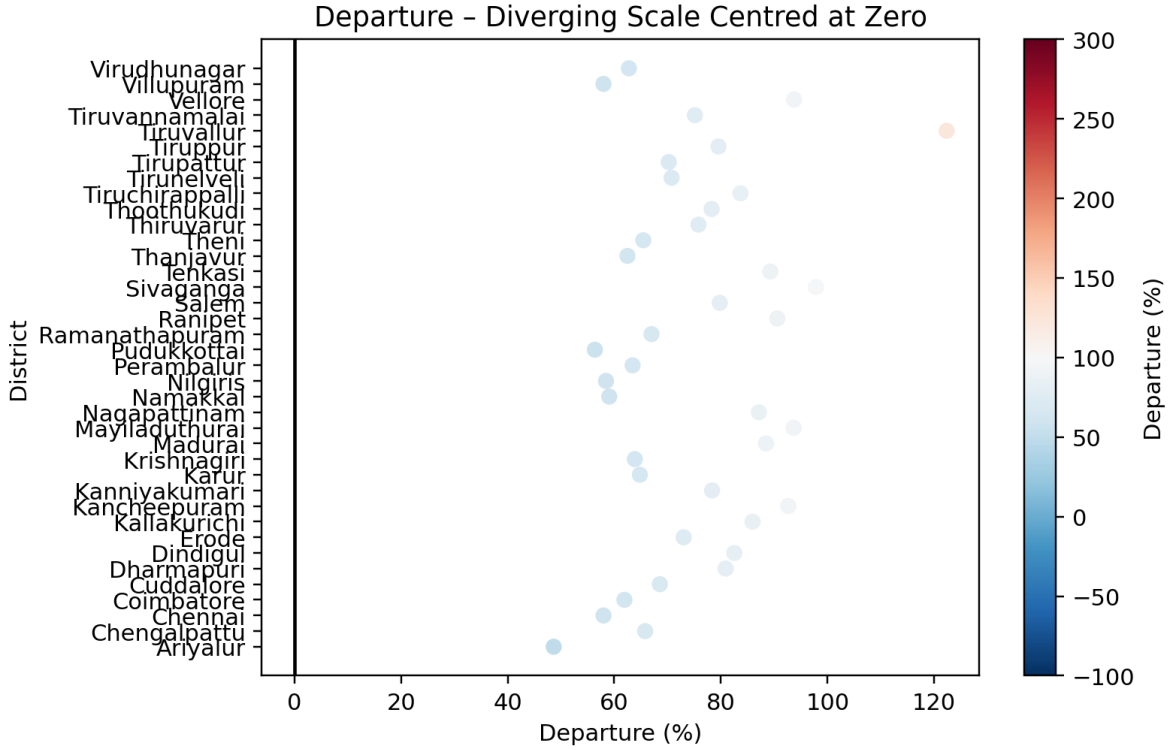


Figure 8. Departure scale centred at zero.

Use blue for negative departure, white at zero and red for positive departure. If midpoint is not zero, normal rainfall may appear colored and mislead. IMD categories must follow Large Deficient → Deficient → Normal → Excess → Large Excess. Highlight coastal districts above +60% in red and all others in grey.

4(c) Palette comparison and Design Decision 3

Palette	Evaluation	Decision
Rainbow / jet	Not perceptually uniform; false boundaries; poor CVD and grayscale behavior.	Reject.
Viridis / plasma	Ordered, perceptually uniform and CVD-friendly; good grayscale progression.	Use for continuous storage.
ColorBrewer	Suitable qualitative and diverging palettes; good for limited categories.	Use Set2 for regions and diverging RdBu-style for departure.

Consistent legend: North=blue, Delta=orange, West=green, South=purple; departure=blue-white-red with zero at white; storage=viridis; coastal alert=red; other districts=grey. A final CVD and grayscale simulator check must be performed and screenshots attached; palette properties alone are not proof of a completed simulation.

Design Decision 3: I recommend viridis for storage, Set2 for four regions and a zero-centred blue-white-red scale for departure. Rainbow is rejected because it is not perceptually uniform and is less accessible.

Stage 5 – Complete Visual Framework (20 marks)
Six required dashboard graphics are provided below.

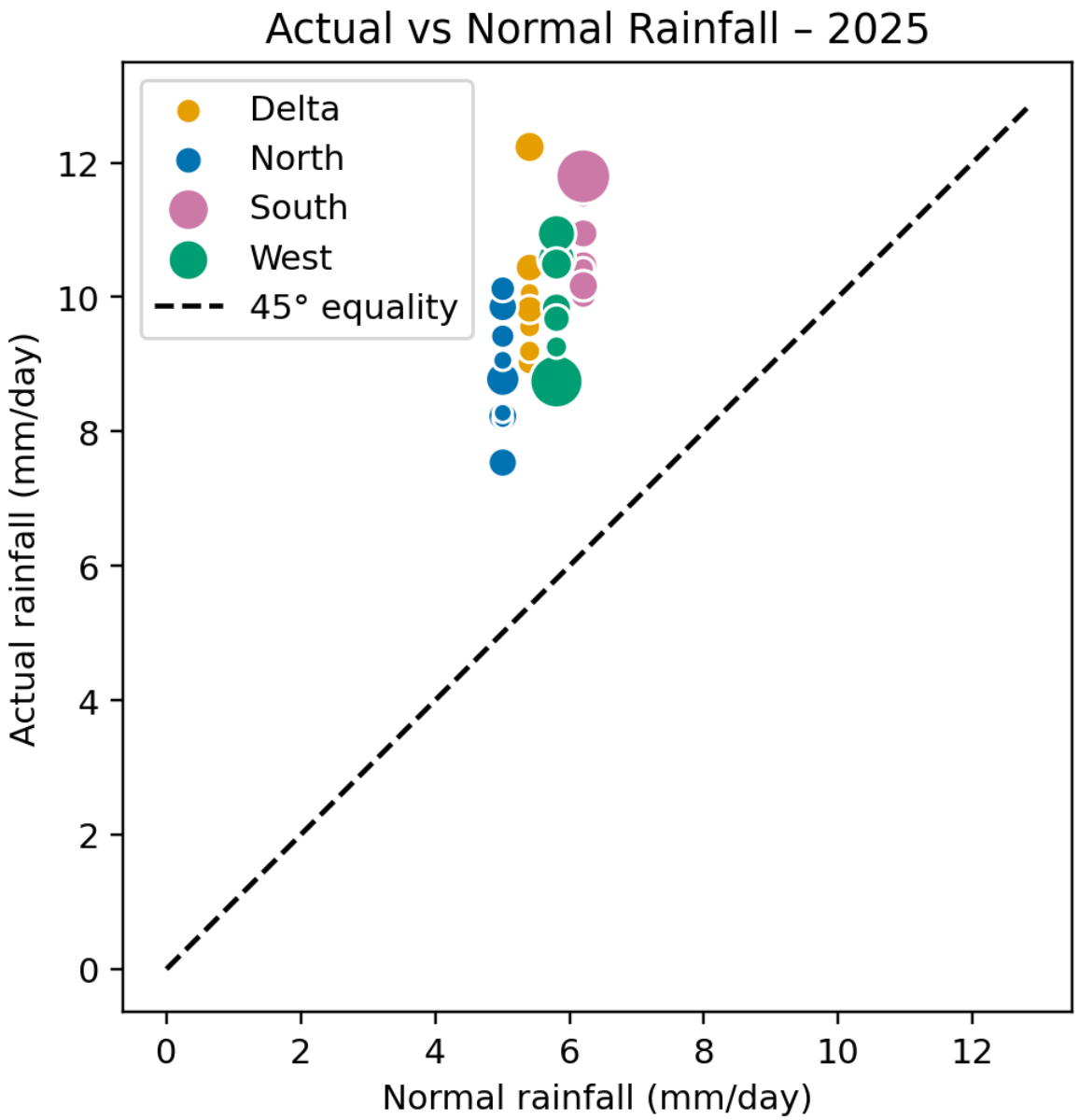


Figure 9. Actual vs normal rainfall scatter with region and evacuation size.

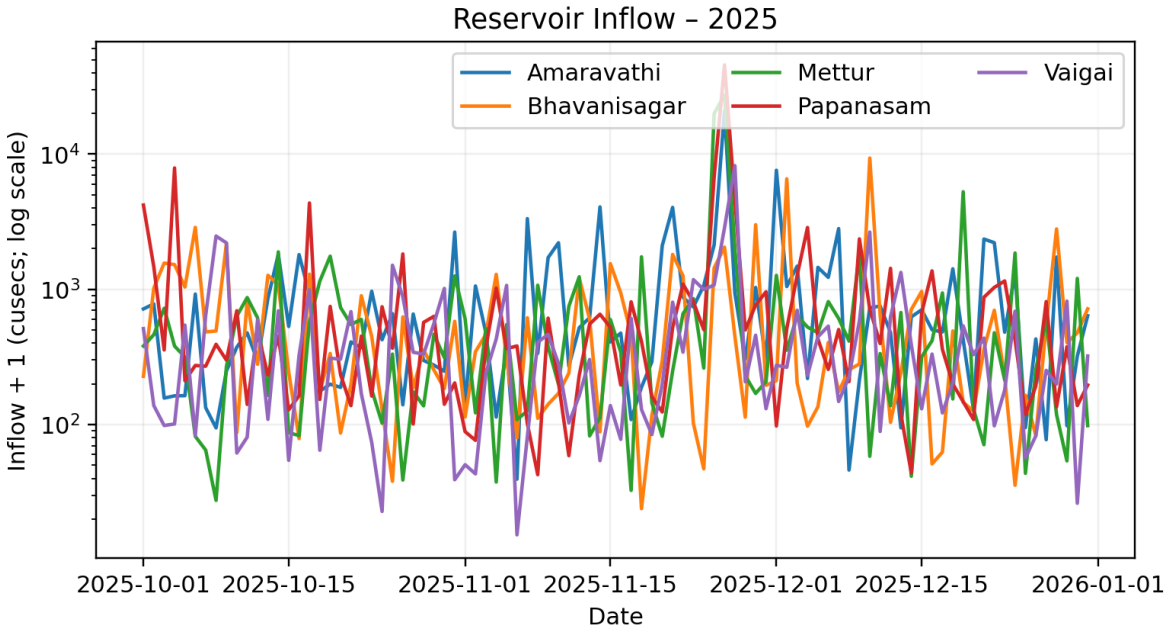


Figure 10. Five major reservoir inflow time series on log scale.

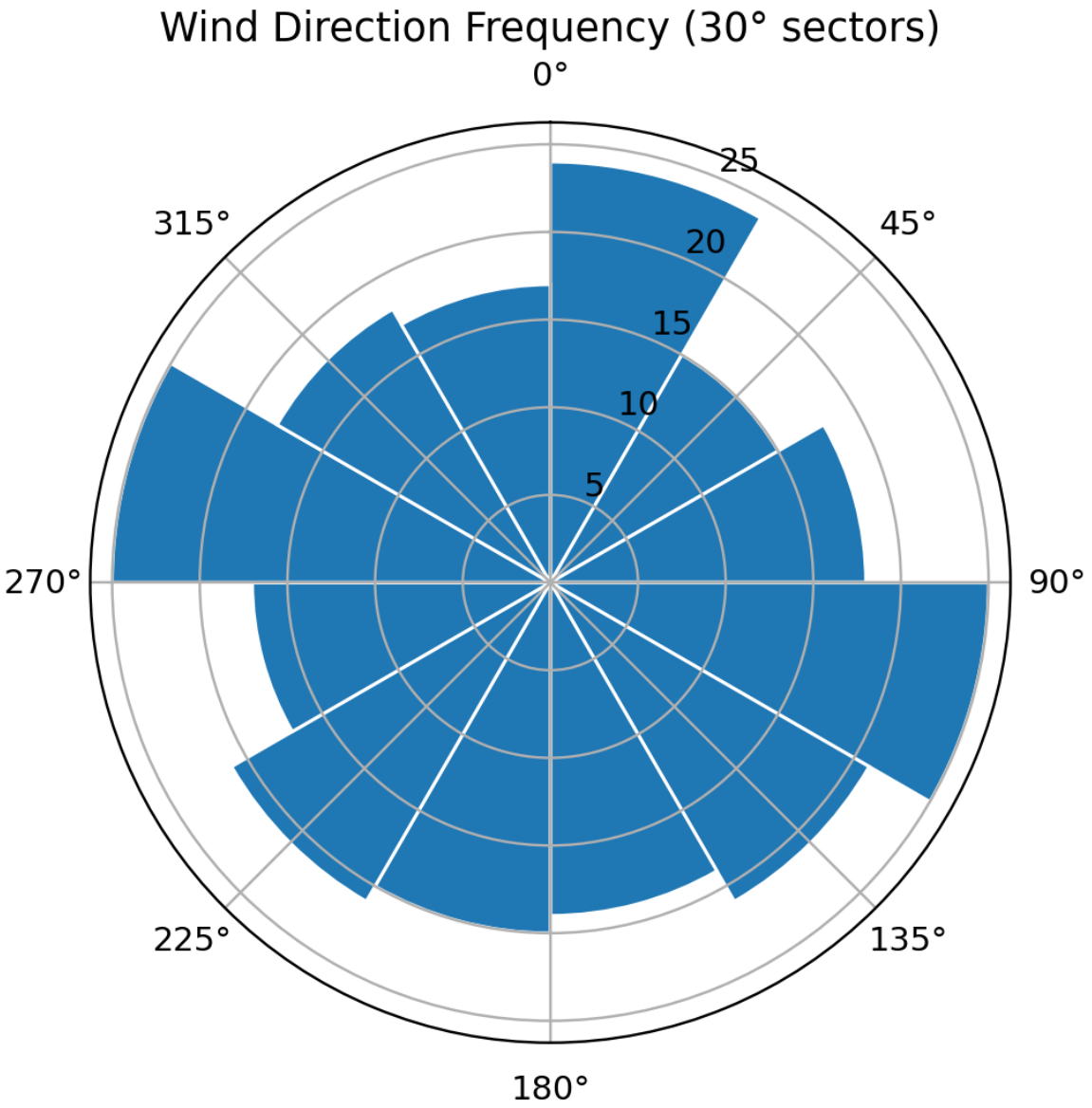


Figure 11. Wind rose.

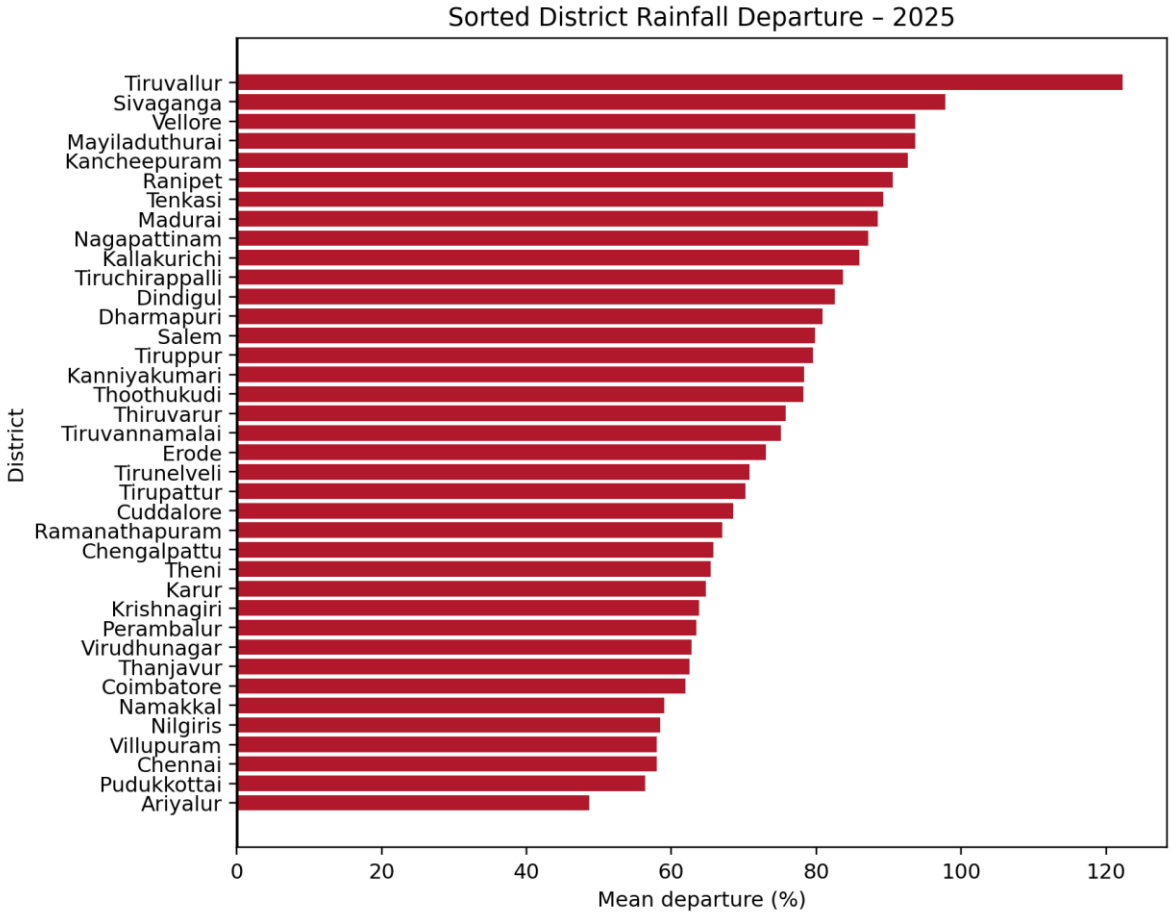


Figure 12. Sorted district departure chart.

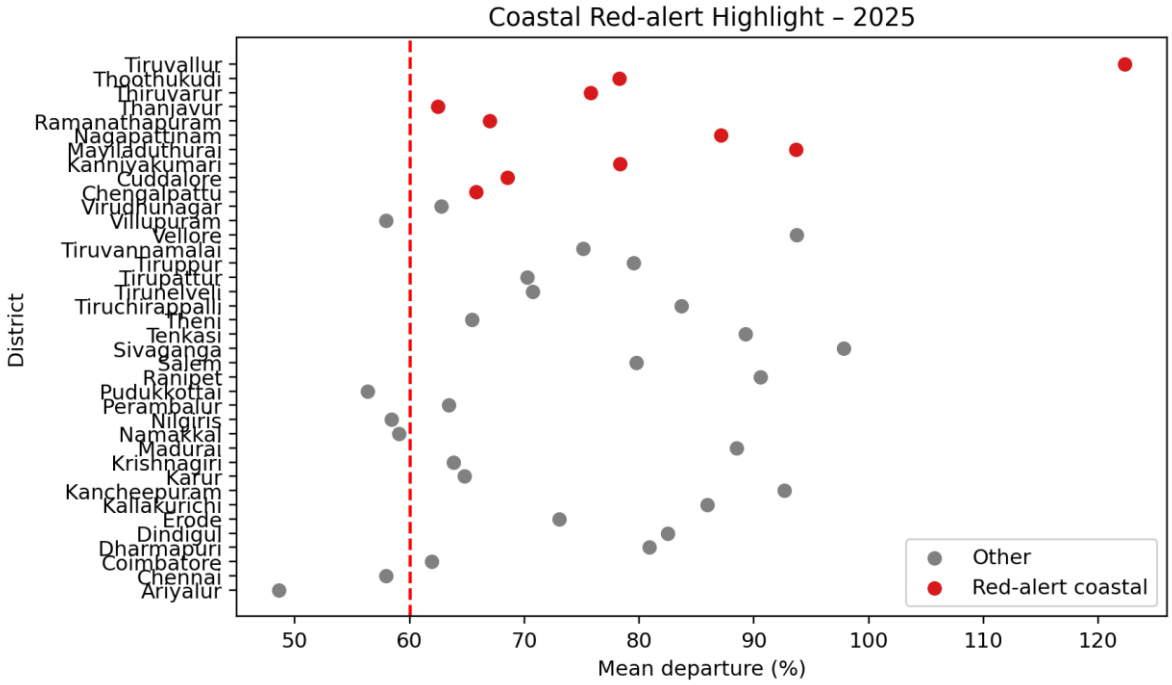


Figure 13. Coastal red-alert highlight.

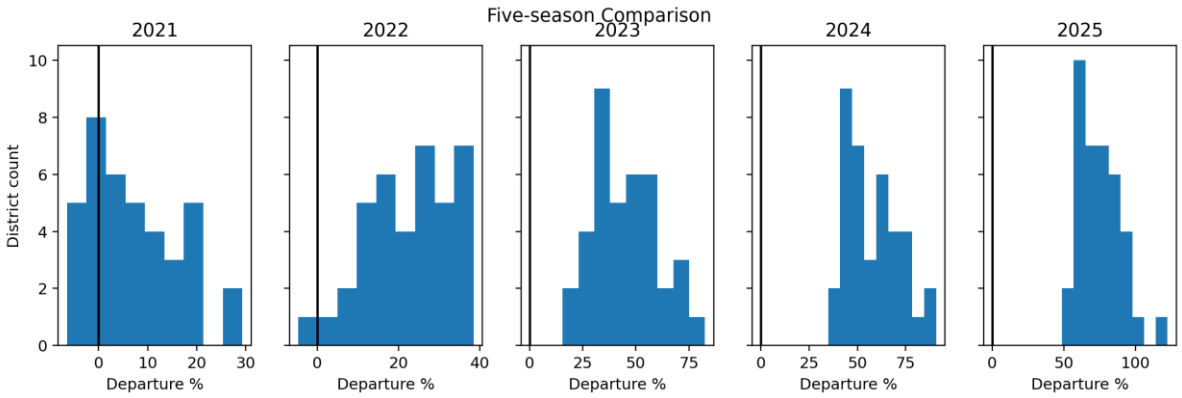


Figure 14. Five-season small multiples with common scales.

5(b) Reusable functions, theme and export

Use a common theme function such as theme_sdma() and reusable functions such as plot_departure(df) and plot_windrose(df). Export each figure with ggsave(..., dpi=300) for PNG and also PDF/SVG for vector quality. To rerun another season, pass the season as a function argument or filter the data by Season. The R code below is a reproducible starter framework; use the official CSVs in place of synthetic data for release.

5(c) Visual Encoding Style Guide

Variable	Aesthetic	Scale	Palette / rule
Region	Color hue	Qualitative	Set2; fixed mapping
Actual/normal	x/y position	Linear, equal aspect	45° equality line
Departure	Position + color	Diverging, midpoint 0	Blue-white-red
IMD category	Fill	Ordered discrete	Deficient to excess
Storage	Color intensity	Sequential continuous	Viridis
Inflow	y position	Log1p; compare linear/sqrt	Label transform
Wind direction	Polar angle	Circular 0°–360°	North at top, clockwise
Evacuated	Point area	Area scale	scale_size_area()
Coastal alert	Accent color	Manual binary	Red alert; grey others

Stage 6 – Verification, Critical Evaluation and Release (20 marks)

6(a) Verification plan

Check	Method	Pass condition
Rainfall scatter	Compare district mean actual/normal with source table.	Coordinates equal source means.
Reservoir inflow	Compare selected dates and inspect log breaks.	Positive order preserved; zero handled with log1p.
Departure	Inspect scale limits and midpoint.	Midpoint exactly 0%.
Point size	Compare counts with symbol areas.	Area proportional to evacuated count.

Accessibility	Export grayscale and run CVD simulator.	Groups remain identifiable with labels/shapes.

Synthetic data checks: 17,480 rainfall rows; normal rainfall is positive; departure is calculated from actual and normal; categories use ordered thresholds; reservoir inflow is non-negative. The demonstration uses inflow + 1 for log plotting. Official data must be rechecked before release.

6(b) Interpretation and critical evaluation

- Positive departure indicates excess rainfall; negative departure indicates deficient rainfall. The sorted chart identifies district extremes.
- The log inflow time series shows ordinary and extreme inflows together; extreme peaks must be confirmed against source records.
- The wind rose identifies frequent direction sectors. Speed-banded sectors are recommended for cyclone intensity assessment.
- The red-alert graphic highlights coastal districts above +60%; this illustrative threshold does not replace official IMD/SDMA alert protocols.
- Small multiples use common scales to make 2021–2025 comparisons fair.

Possible misleading issue	Risk	Control
Truncated axis	Exaggerates differences.	Use meaningful baselines and label zooms.
Log axis for public	Equal visual steps may be misunderstood.	Explain transformation and provide linear companion.
Radius vs area	Overstates large counts.	Use area-scaled points.
Pie with many slices	Angles are hard to compare.	Prefer ordered bar chart.
Too many colors	Groups become indistinguishable.	Use facets, labels and grouping.
Color-only alert	CVD/grayscale viewers may miss it.	Add alert labels and symbols.

6(c) Requirement verification table

Requirement	Specified	Achieved in report	Status
Variable classification	All variables	Classification table	YES
Aesthetic mapping	Suitable mappings	Three alternatives and rationale	YES
Scales and breaks	Meaningful scales	Date, log1p and diverging explanations	YES
Zoom vs filtering	Explain difference	coord_cartesian explanation	YES
Nonlinear axis	Log and zero handling	Linear/log/sqrt comparison	YES
Polar coordinates	Wind rose and pie	Wind rose and pie explanation	YES
Color to distinguish	Four regions	Qualitative palette	YES
Color to represent values	Sequential/diverging	Viridis and zero-centred scale	YES

Color to highlight	Coastal alerts	Red/grey graphic	YES
CVD / grayscale	Tests and screenshots	Palette rationale; screenshots still needed	PARTIAL
Consistent legend	Same meaning throughout	Style guide	YES
Visual questions answered	All SDMA questions	Dashboard and interpretation	YES

Final Visualization Decision

DECISION: NO – not yet for operational release. The design is suitable as a prototype, but all data in this report are synthetic, not official IMD/SDMA observations. In addition, grayscale and CVD simulator screenshots must be attached and operational alert thresholds must be approved.

Before release: replace synthetic data with validated official records; check units, missing values, district IDs and dates; compare at least three plotted values against source tables; attach grayscale and CVD screenshots; confirm alert thresholds with SDMA/IMD; test mobile and black-and-white output; and complete all verification checks. These steps improve accuracy, accessibility, reliability and public safety.

Appendix A – R Code: Reproducible Core and Plotting Templates

```
# DSA06 CO1 - R / ggplot2 core templates
# Replace synthetic input with official faculty / IMD CSV files.
library(ggplot2)
library(dplyr)
library(scales)
library(viridis)

# Read the supplied synthetic dataset or replace with official data
rain <- read.csv("synthetic_rainfall_2021_2025.csv")
rain$Date <- as.Date(rain$Date)
rain$Region <- factor(rain$Region,
  levels=c("North","Delta","West","South"))
rain$IMD_Category <- factor(rain$IMD_Category,
  levels=c("Large Deficient","Deficient","Normal","Excess","Large Excess"),
  ordered=TRUE)

theme_sdma <- function() {
  theme_minimal(base_size=11) +
    theme(plot.title=element_text(face="bold"),
          legend.position="bottom",
          panel.grid.minor=element_blank())
}

# District summary for selected season
district_summary <- function(df, season=2025) {
```

```
  df %>% filter(Season==season) %>%
    group_by(District,Region,Coastal) %>%
    summarise(Actual=mean(Actual_mm),
              Normal=mean(Normal_mm),
              Departure=mean(Departure_pct),.groups="drop")
}

# 1. Scatter: region color, 45-degree line, equal aspect
plot_rain_scatter <- function(df, season=2025) {
  d <- district_summary(df,season)
  ggplot(d,aes(Normal,Actual,color=Region)) +
    geom_point(size=3) +
    geom_abline(slope=1,intercept=0,linetype=2) +
    coord_fixed() + scale_color_brewer(palette="Set2") +
    labs(title=paste("Actual vs normal rainfall",season),
         x="Normal rainfall (mm/day)",y="Actual rainfall (mm/day)",
         color="Region") + theme_sdma()
}

# 2. Daily state-average rainfall and zoom
daily <- rain %>% filter(Season==2025) %>%
  group_by(Date) %>% summarise(StateMean=mean(Actual_mm),.groups="drop")
p_daily <- ggplot(daily,aes(Date,StateMean)) + geom_line() +
  scale_x_date(date_breaks="2 weeks",date_labels="%d %b") +
  labs(title="Daily state-average rainfall",
       x="Date",y="Mean rainfall (mm/day)") + theme_sdma()
p_zoom <- p_daily +
  coord_cartesian(xlim=as.Date(c("2025-11-20","2025-12-05")))

# 3. Reservoir inflow: zero-safe log1p; linear and sqrt alternatives
reservoir <- read.csv("synthetic_reservoirs.csv")
reservoir$Date <- as.Date(reservoir$Date)
p_inflow_log <- ggplot(subset(reservoir,Season==2025),
  aes(Date,Inflow_cusecs+1,color=Reservoir)) +
  geom_line() + scale_y_log10() +
  labs(title="Reservoir inflow (log10 of inflow + 1)",
       x="Date",y="Inflow + 1 (cusecs)",color="Reservoir") +
  theme_sdma()
p_inflow_linear <- ggplot(subset(reservoir,Season==2025),
  aes(Date,Inflow_cusecs,color=Reservoir)) + geom_line() +
  labs(title="Reservoir inflow – linear",y="Inflow (cusecs)") +
  theme_sdma()
p_inflow_sqrt <- ggplot(subset(reservoir,Season==2025),
  aes(Date,sqrt(Inflow_cusecs),color=Reservoir)) + geom_line() +
```

```
  labs(title="Reservoir inflow – square root",y="sqrt(inflow)") +
  theme_sdma()

# 4. Wind rose from direction sectors
wind <- read.csv("synthetic_wind.csv")
wind$Sector <- cut(wind$Direction_deg,breaks=seq(0,360,by=30),
                   include.lowest=TRUE,right=FALSE)
wind_counts <- as.data.frame(table(wind$Sector))
p_windrose <- ggplot(wind_counts,aes(Sector,Freq,fill=Freq)) +
  geom_col(width=1,color="white") + coord_polar() +
  scale_fill_viridis_c() +
  labs(title="Wind rose – direction frequency",
       x="Direction sector",y="Frequency",fill="Frequency") +
  theme_sdma()

# 5. Departure: diverging scale centred exactly at zero
plot_departure <- function(df,season=2025) {
  d <- district_summary(df,season)
  ggplot(d,aes(Departure,reorder(District,Departure),color=Departure)) +
    geom_point(size=3) +
    scale_color_gradient2(low="#2166AC",mid="white",high="#B2182B",
                          midpoint=0) +
    labs(title=paste("Rainfall departure",season),
         x="Departure (%)",y="District",color="Departure (%)") +
    theme_sdma()
}

# 6. Highlight coastal districts above +60 percent
plot_alert <- function(df,season=2025) {
  d <- district_summary(df,season)
  d$Alert <- ifelse(d$Coastal & d$Departure>60,
                    "Red-alert coastal","Other districts")
  ggplot(d,aes(Departure,reorder(District,Departure),color=Alert)) +
    geom_point(size=3) +
    geom_vline(xintercept=60,linetype=2,color="red") +
    scale_color_manual(values=c("Red-alert coastal"="#D7191C",
                                "Other districts"="grey70")) +
    labs(title="Coastal red-alert highlight",
         x="Departure (%)",y="District",color="Alert") + theme_sdma()
}

# 7. Storage sequential color
p_storage <- ggplot(subset(reservoir,Season==2025),
  aes(Date,Reservoir,color=Storage_pct)) +
```

```
  geom_point() + scale_color_viridis_c() +
  labs(title="Reservoir storage",x="Date",y="Reservoir",
       color="Storage (%)") + theme_sdma()

# 8. IMD ordered categories
p_imd <- ggplot(subset(rain,Season==2025),
  aes(IMD_Category,fill=IMD_Category)) +
  geom_bar() + scale_fill_brewer(palette="RdYlBu",direction=-1) +
  labs(title="Ordered IMD rainfall categories",
       x="IMD category",y="District-day count") + theme_sdma()

# 9. Area-scaled evacuation size
evac <- read.csv("synthetic_evacuation.csv")
d <- district_summary(rain,2025) %>% left_join(evac,by="District")
p_evac <- ggplot(d,aes(Normal,Actual,size=Evacuated,color=Region)) +
  geom_point(alpha=.8) + scale_size_area(max_size=12) +
  scale_color_brewer(palette="Set2") +
  labs(title="Rainfall and evacuated population",
       x="Normal rainfall (mm/day)",y="Actual rainfall (mm/day)",
       size="People evacuated",color="Region") + theme_sdma()

# Export PNG 300 dpi and vector formats
ggsave("rain_scatter.png",plot_rain_scatter(rain),width=8,height=6,dpi=300)
ggsave("rain_scatter.pdf",plot_rain_scatter(rain),width=8,height=6)
ggsave("rain_scatter.svg",plot_rain_scatter(rain),width=8,height=6)
ggsave("daily_rainfall.png",p_daily,width=8,height=5,dpi=300)
ggsave("rain_zoom.png",p_zoom,width=8,height=5,dpi=300)
ggsave("inflow_log.png",p_inflow_log,width=8,height=5,dpi=300)
ggsave("windrose.png",p_windrose,width=7,height=6,dpi=300)
ggsave("departure.png",plot_departure(rain),width=8,height=7,dpi=300)
ggsave("alert.png",plot_alert(rain),width=8,height=7,dpi=300)

# Verification checks
stopifnot(nrow(rain)==17480)
stopifnot(all(rain$Normal_mm>0))
stopifnot(all(is.finite(rain$Departure_pct)))
stopifnot(all(reservoir$Inflow_cusecs>=0))
stopifnot(min(c(-100,0,300))== -100) # signed departure includes zero
cat("Verification checks passed. Log uses inflow + 1; diverging midpoint is 0.\\n")
```

Appendix B – Data and evidence files

The synthetic CSV datasets are supplied separately: synthetic_rainfall_2021_2025.csv, synthetic_reservoirs.csv, synthetic_wind.csv and synthetic_evacuation.csv. The embedded figures are generated from these illustrative values. The assignment requires official data, exported charts, code and verification evidence; replace the demonstration data and attach actual grayscale/CVD screenshots before final operational submission.